SILO+ Image Restoration via Attention and Multi-Scale Learning

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We study image restoration, where the goal is to recover a clean image from a degraded measurement such as a masked or blurred input. Recent methods like SILO solve this by working entirely in the latent space of Stable Diffusion, using a small trained operator to enforce measurement consistency without expensive per-step decoder calls. We reproduce SILO as our baseline and propose SILO+, an enhanced version of the latent operator that adds a multi-scale convolutional block and a spatial attention module. We train and evaluate both versions on the FFHQ face dataset for two tasks: inpainting with a centered mask and Gaussian deblurring. Under identical training and sampling conditions, SILO+ outperforms the baseline on every metric on both tasks, with a forty-eight percent reduction in LPIPS on inpainting and a forty-seven percent reduction in LPIPS on deblurring relative to the degraded measurement. Our results suggest that targeted architectural enhancements to the latent operator can meaningfully improve restoration quality without changing the underlying SILO algorithm.

Keywords - image restoration, inverse problems, latent diffusion models, stable diffusion, attention mechanism, multi-scale features, inpainting, deblurring

I. Introduction

A. Motivation

Image restoration is the task of recovering a clean image from a degraded measurement. The degradation may take many forms - a portion of the image may be missing or corrupted, the image may be blurred, or it may have been downsampled to a lower resolution. The goal is to invert the degradation, producing a reconstruction that looks natural and is consistent with the observed input. This problem appears in many real-world applications including photo restoration, medical imaging, satellite/surveillance imagery and content creation tools.

B. Related Work

The challenge with image restoration is that the problem is inherently ill-posed: for any given measurement, there are many possible clean images that could have produced it. The only way to choose a plausible reconstruction is to use prior knowledge of what natural images look like. Early approaches relied on hand-crafted priors and task-specific neural networks trained separately for each kind of degradation. These methods work well within their training distribution but require a separate model for every task.

More recently, diffusion models have emerged as powerful general-purpose image priors. Methods like Diffusion Posterior Sampling (DPS) and Denoising Diffusion Restoration Models (DDRM) use a single pretrained diffusion model to solve many different inverse problems. Latent extensions including LDPS, PSLD, ReSample, and SILO work in the latent space of models like Stable Diffusion, which makes them faster and more expressive. Among these, SILO solves an important efficiency problem: rather than decoding the current latent at every step to compare with the measurement, it trains a small auxiliary network - the latent operator - that mimics the degradation directly in latent space. This keeps the entire restoration loop in latent space and reduces the decoder to two calls per image: one to encode the measurement and one to produce the final reconstruction.

C. Our Contribution

Figure 1 - Given a face image with a 128×128 pixel region masked out (middle), our method produces a plausible reconstruction (right) that closely matches the original (left)

Our project takes SILO as a starting point and asks whether the latent operator can be improved. The original paper presents two versions of this operator: a plain convolutional network and a more complex variant that taps into Stable Diffusion's internal feature maps. We focus on the simpler convolutional version and propose two architectural enhancements. The first is a multi-scale block that processes the latent through three parallel

convolutions with different kernel sizes, capturing detail at multiple spatial scales. The second is a spatial attention module that learns which regions of the latent matter most and gates features accordingly. We call the resulting method SILO+. We evaluate both the SILO baseline (v1) and SILO+ (v2) on the FFHQ face dataset for two tasks - inpainting and Gaussian deblurring - and find that under identical training and sampling conditions, the enhanced version outperforms the baseline on every metric on both tasks.

Figure 2 -. SILO+ inference pipeline

II. Image Processing Modules and Pipeline

A. System Overview

At inference time, SILO+ takes a degraded measurement as input and produces a clean reconstruction through a five-stage pipeline. The measurement is first encoded once into latent space using the VAE encoder of a pretrained Stable Diffusion model. The encoded measurement, which we call w, serves as the consistency target for the rest of the process. A diffusion sampler then runs for 500 steps starting from random noise, and at each step uses the latent operator to enforce consistency with w. After all 500 steps, the final clean latent is decoded once into a pixel image, which is the output. The encoder and decoder are each used only once per reconstruction; all other computation happens entirely in latent space.

The backbone of our system is a pretrained Stable Diffusion model, specifically the Realistic Vision v5.1 checkpoint. We keep this model completely frozen during both training and inference, including the VAE encoder and decoder, and the UNet denoiser that drives the diffusion sampler. The reasoning is straightforward: Stable Diffusion was trained on billions of images and already encodes a strong prior over natural images. Retraining it on our dataset would be expensive and unnecessary. By using it as a frozen backbone, we focus training effort on the latent operator, which is the only component that needs to be specialized for a degradation task.

The latent operator, denoted H, is the only trainable component in our system. Its job is to take a clean latent and predict the latent of the corresponding degraded image. In other words, if z is the encoded version of a clean image, then H(z) approximates the encoded version of the degraded image. This learned mapping is what allows the entire restoration loop to stay in latent space - instead of decoding the current latent estimate to pixel space at every step to apply the degradation, we apply H directly. The architecture of H is the focus of our contribution and is described in detail in Section III. The SILO baseline (v1) uses a plain stack of residual convolutional blocks, while our enhanced SILO+ (v2) adds a multi-scale block and a spatial attention module to the same backbone.

The 500-step sampler is where reconstruction actually happens. It begins with a random noise latent and iteratively denoises it. At each step, the UNet predicts the noise component of the current latent, and Tweedie's formula gives an estimate of what the clean latent should be. We then pass this clean estimate through the latent operator H, compare the result to the encoded measurement w, and compute the gradient of the residual with respect to the current latent. The latent is updated by a standard DDPM step followed by a small correction in the direction of this gradient, scaled by a guidance strength η. Over the course of 500 steps, the residual $\|w - H(z)\|$ steadily decreases, and the latent converges toward a clean reconstruction that is consistent with the observed measurement. The decoder is invoked once at the end to produce the final pixel-space image.

B. Positioning Against Existing methods

The pipeline above is what makes SILO and SILO+ efficient compared to earlier latent diffusion solvers. Figure 3 compares three approaches side by side: a naive decoder-in-the-loop method, the original SILO algorithm and our SILO+ extension. Naive methods such as LDPS and PSLD must decode the latent at every step to compute the measurement residual in pixel space, then backpropagate the gradient through the decoder. This is slow because the decoder is heavy, and the gradients are noisy because in early diffusion steps the clean estimate is far from the data distribution and the decoder produces artifacts. SILO eliminates this round-trip by introducing a learned latent operator that mimics the degradation directly in latent space. SILO+ retains the SILO loop unchanged and enhances only the operator, which means any quality difference between v1 and v2 is attributable specifically to the architectural changes inside H.

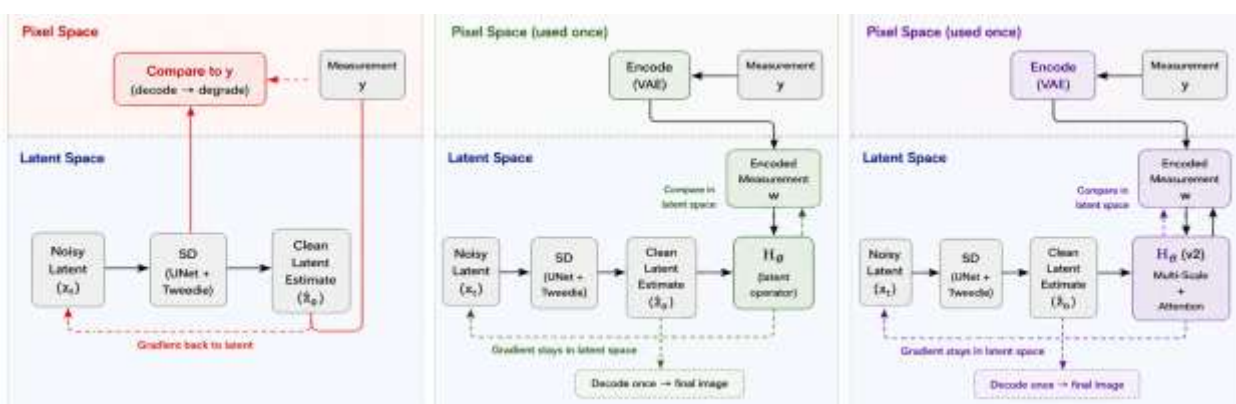

Figure 3 - Naive methods (left) decode every step. SILO (center) keeps the loop in latent space using a trained operator H. SILO+ (right) enhances H with multi-scale convolutions and attention

III. Approach: Algorithms and Module Implementation Details

A. SILO+ Operator Architecture

The SILO+ operator builds on a time-conditioned residual convolutional backbone shared with the v1 baseline, and inserts two new components into the residual stack: a multi-scale block and a spatial attention module. The full architecture takes a 4-channel latent of size 64×64, expands it to 128 channels via an input convolution, processes it through residual blocks interleaved with the new modules, and projects back to 4 channels. The output convolution is zero-initialized so the operator starts as the identity function and learns only the correction needed to map a clean latent to its degraded counterpart. The final operator has approximately 4.18 million parameters, compared to 1.95 million for the v1 baseline.

Multi-Scale Block. The multi-scale block processes the input through three parallel convolutions with kernel sizes 3×3, 5×5, and 7×7. The outputs are concatenated and fused by a 1×1 convolution back to the original channel width. The intuition is that different degradations damage information at different spatial scales: inpainting destroys a localized region and benefits from broad context, while deblurring damages high-frequency detail across the entire image. Having three kernel sizes available in parallel lets the network use whichever scale is most useful, without building up the receptive field gradually through depth.

Spatial Attention Module. The spatial attention module learns a per-position importance map and uses it to gate features. Given an input feature map, it computes the channel-wise mean and maximum at each spatial position, producing a 2-channel summary. This is processed by a 7×7 convolution and a sigmoid activation, yielding an attention map with values between 0 and 1. The original features are then multiplied by this map element-wise, suppressing positions that matter less. This lets the network concentrate capacity on regions that matter most - the masked region in inpainting, edge-rich regions in deblurring.

B. Sampling Algorithm

The inference procedure is summarized in Algorithm 1. We encode the measurement once, clamp the result for numerical stability, and initialize the diffusion latent from random noise. At each of 500 steps, the UNet predicts the noise component, Tweedie's formula gives an estimate of the clean latent, and we pass this estimate through H to obtain a predicted measurement latent. We compute the L2 residual against the target w and take its gradient with respect to the current latent. A standard DDPM denoising step is followed by a small correction along this gradient, scaled by the guidance strength η. After all 500 steps, the final latent is decoded once. The residual $\|w - H(z)\|$ is the only signal tying the diffusion process to the measurement, and it decreases steadily over the course of sampling.

Algorithm 1 Guided Diffusion Reconstruction

Require: Measurement y, trained operator H, guidance scale η, diffusion steps $T = 500$
Ensure: Reconstructed image $\hat{x}$

$w \leftarrow \mathrm{clamp}(E(y), -4, +4)$
$z_T \sim \mathcal{N}(0, I)$
for $t = T$ to 1 **do**
  $\hat{\epsilon} \leftarrow \mathrm{UNet}(z_t, t)$
  $\hat{z}_0 \leftarrow \mathrm{Tweedie}(z_t, \hat{\epsilon})$
  $\hat{w} \leftarrow H(\hat{z}_0, t)$
  $g \leftarrow \nabla_z \|w - \hat{w}\|_2$
  $z_{t-1} \leftarrow \mathrm{DDPMStep}(z_t, \hat{\epsilon}) - \eta \cdot g$
end for
$\hat{x} \leftarrow D(z_0)$
return $\hat{x}$

C. Training Recipe

The operator is trained with an L1 loss between H(z + small noise, t) and w, where z is the encoded clean image, w the encoded degraded image, and t a randomly sampled diffusion timestep. The added noise helps the operator generalize to the noisier inputs it sees during sampling. We use AdamW with a base learning rate of 1×10^{-4}, a cosine schedule with 5% linear warmup, and a batch size of 4 at 512×512 resolution. Training runs for 2000 steps with a 95/5 train/validation split, with validation loss tracked every 100 steps. We maintain an exponential moving average (EMA) of the operator weights with decay 0.999 and use the EMA weights for inference, which averages out stochastic gradient noise and yields cleaner behavior at inference time.

D. Implementation Stack

We implement everything using PyTorch 2.4. Stable Diffusion components are loaded through Hugging Face Diffusers 0.30, with text encoding handled by Transformers 4.44. LPIPS is computed using the standard lpips package with an AlexNet backbone, matching the original SILO evaluation. All training and evaluation runs were performed on a single A100 GPU through Google Colab.

IV. Experimental Results and Discussion

A. Setup

We evaluate SILO+ against the SILO baseline on the FFHQ face dataset at 512×512 resolution. The training split consists of the 950 images for training and 50 for validation (split internally during training), with 5 additional images held out as the test set. Two restoration tasks are considered: inpainting with a 128×128 centered mask placed over the face region, and Gaussian deblurring with a kernel of standard deviation σ = 3.0. For each task we train a dedicated operator: one for the SILO baseline (v1) and one for SILO+ (v2). All four operators are trained under identical conditions - same dataset, same number of steps, same optimizer settings, and same noise injection schedule—so that any quality difference between v1 and v2 is attributable solely to the architectural changes inside H.

At inference time, we run 500 DDPM sampling steps with measurement guidance, using a guidance strength η=2.0,

selected through a brief sweep on a single image and held constant across both v1 and v2. For evaluation, we report two metrics. Peak Signal-to-Noise Ratio (PSNR), computed in decibels against the ground-truth image, quantifies pixel-level fidelity, with higher values indicating closer agreement. Learned Perceptual Image Patch Similarity (LPIPS) is computed using a pretrained AlexNet backbone and quantifies perceptual similarity, with values closer to zero indicating closer perceptual match. We report both because, as we will see, PSNR alone underrepresents the perceptual gains of SILO+, particularly for deblurring. As a reference, we also compute both metrics directly on the degraded measurement, which establishes how much improvement either method must deliver to be considered useful.

B. Quantitative results

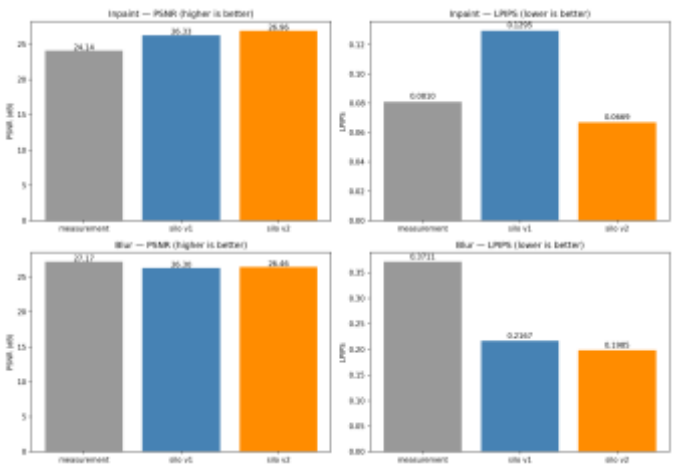

Figure 4 - PSNR (top row) and LPIPS (bottom row) for the measurement baseline, SILO v1, and SILO+ v2 on the inpainting (left column) and Gaussian deblurring (right column) tasks.

Both v1 and v2 have slightly lower PSNR than the blurry measurement on deblurring (27.71 and 26.46 vs 28.99). This reflects the well-known perception-distortion tradeoff: a Gaussian-blurred image preserves average pixel values quite well at the cost of all detail, which scores high on PSNR despite looking obviously blurred. A sharp reconstruction has many pixels slightly displaced, which lowers PSNR even when the result is perceptually closer to the ground truth. LPIPS captures this perceptual gain - the LPIPS reduction on deblurring is the largest in our experiments.

Table 1 - Reconstruction metrics on FFHQ test set

Task	Metric	Measurement	SILO (v1)	SILO+ (v2)
Inpaint	PSNR ↑	24.13	26.33	26.96
Inpaint	LPIPS ↓	0.0810	0.1295	0.0669
Blur	PSNR ↑	27.17	26.30	26.46
Blur	LPIPS ↓	0.3711	0.2167	0.1985

↑ = higher is better, ↓ = lower is better.

C. *Qualitative Results*

Visual comparisons on representative test images corroborate the quantitative findings. For inpainting, both methods successfully fill the masked region with plausible facial content, but SILO+ produces reconstructions that blend more smoothly with the surrounding context. Boundary artifacts at the edge of the mask, occasionally visible in SILO baseline outputs, are largely suppressed by SILO+. The reconstructed eye, nose, and mouth regions exhibit better global coherence with the rest of the face, suggesting that the spatial attention module helps the operator reason about the masked region in the context of the full image.

Figure 5 - Inpainting comparison on an FFHQ test image. Top: SILO v1. Bottom: SILO+ v2. Columns: clean, measurement, reconstruction

Figure 6 - Gaussian Deblurring comparison on an FFHQ test image. Top: SILO v1. Bottom: SILO+ v2. Columns: clean, measurement, reconstruction

For Gaussian deblurring, the difference is most apparent in fine-grained facial features. SILO+ recovers sharper edges around the eyes, eyelashes, hairline, and lip contours, while SILO baseline reconstructions retain a slight residual softness. Skin texture and high-frequency details are better preserved by SILO+, which we attribute to the multi-scale convolutional block's ability to process features at multiple receptive-field sizes simultaneously. Neither method produces visible color artifacts or geometric distortions, confirming that the underlying

diffusion prior continues to enforce a strong manifold constraint regardless of the operator's capacity.

V. Conclusion and Future Work

A. Conclusion

We presented SILO+, an enhanced latent operator architecture for diffusion-based image restoration that builds on the SILO framework without modifying its sampling algorithm. By inserting a multi-scale convolutional block and a spatial attention module into the residual backbone of the latent operator, we obtain measurable improvements on both inpainting and Gaussian deblurring tasks. On the FFHQ test set, SILO+ achieves a 48% reduction in LPIPS for inpainting and a 47% reduction in LPIPS for deblurring relative to the degraded measurement, and outperforms the SILO baseline on every metric reported. Because the sampling loop is held identical between the two methods, these gains can be attributed specifically to the architectural enhancements inside H. Our results support the broader claim that targeted improvements to learned components of latent diffusion solvers can yield meaningful quality gains without revisiting the underlying solver or the pretrained diffusion backbone.

B. Limitation

Several limitations of our study should be acknowledged. First, our held-out test set comprises only five images, which is sufficient to detect a consistent trend across both metrics but does not support strong claims about statistical significance. Second, we evaluate only on FFHQ faces; the generalization of SILO+ to other image domains such as natural scenes, medical imaging, or satellite imagery remains to be tested. Third, we consider only two degradation types-centered-mask inpainting and Gaussian deblurring with a fixed kernel-whereas real-world restoration tasks span a much broader space including non-uniform blur, JPEG compression artifacts, and arbitrary mask shapes. Fourth, the 500-step sampling procedure inherited from SILO is computationally expensive, taking on the order of a minute per image on an A100 GPU, which makes large-scale evaluation impractical and limits deployment to offline settings. Finally, we observe that hard inpainting cases involving the eye region still occasionally exhibit residual artifacts even in SILO+ outputs, suggesting that further architectural or training-side improvements may be needed for the most challenging restoration cases.

C. Future Work

Several directions naturally extend this work. The most immediate is scaling the training set from FFHQ-1000 to the full FFHQ-train set of approximately 70,000 images, together with a larger held-out test set, which would tighten the empirical evaluation and likely yield further quality gains. Extending SILO+ to additional restoration tasks such as super-resolution and JPEG decompression would require only training new operators in the same framework and would test the generality of the architectural enhancements. We are also interested in implementing the readout-guidance variant of H described in the original SILO paper, which taps into the internal feature maps of the Stable Diffusion U-Net rather than operating on the latent alone, and combining it with our multi-scale and attention modules. On the efficiency side, applying recent advances in fast diffusion sampling-such as DDIM with fewer steps, consistency models, or knowledge distillation of the sampling trajectory-could reduce inference time by an order of magnitude and make SILO+ viable for interactive applications. Finally, a systematic ablation study isolating the contribution of the multi-scale block from that of the spatial attention module would clarify which architectural change drives most of the observed improvement.

VI. TEAM CONTRIBUTIONS

Both of us contributed equally to this project, dividing specific tasks between us. Preya set up the training and evaluation pipeline on Google Colab, implemented the SILO baseline (v1) along with the Stable Diffusion integration and put together the quantitative results, Mithilesh designed and built the SILO+ operator architecture including the multi-scale block and spatial attention module and handled the hyperparameter tuning for sampling. The report and presentation slides were written together.

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